Raising the temperature ratio limit for establishing phonon antibunching by coupling an array of Carbon nanotubes

A THESIS

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DECLARATION BY STUDENT

I, Jayant Bakolia, hereby declare that the work presented herein is original work done by me and has not been published or submitted elsewhere for the requirement of a degree programme. Any literature date or work done by others are cited within this project has given due acknowledgment and listed in the reference section.

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CERTIFICATE

This is to certify that the thesis entitled **“**Raising temperature ratio limit for establishing phonon antibunching by coupling an array of Carbon nanotubes**”** submitted by **JAYANT BAKOLIA** to the Indian Institute of Technology, Madras, for the award of the degree of **Master of Technology in Automotive Technology** is a bonafide record of project work carried out by him under my supervision. The contents of this thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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ABSTRACT

In this work, we explore the coupling of an array of phonon modes to achieve a higher limit on the temperature ratio (i.e., the ratio of surrounding temperature to characteristic system temperature due to vibrations) concerning the second-order correlation coefficient. We propose a driven-dissipative nanomechanical system wherein an array of coupled Carbon nanotubes is used as resonators. Furthermore, this project uses non-local analytical concepts to find suitable system parameters for realizing the phonon blockade effect at around 1 Kelvin range.

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NOMENCLATURE

$a^\dagger$	Conjugate transpose of a
$g^{(2)}(0)$	Second order correlation coefficient
F	Amplitude of force through which resonator is driven
J	Coupling strength
U	Kerr nonlinearity
U^e	Potential energy
Δ	Detuning of the natural frequency with driven frequency
$\hat{\rho}$	Reduced density matrix
$\hat{\rho}_{ss}$	Steady-state reduced density matrix
$\otimes$	Tensor product
A^H	Conjugate transpose of A
ι	iota
b_i	Annihilation operator
T_0	System characteristic temperature
T	Surrounding temperature or Operating temperature
h	Planck's constant
K_B	Boltzmann constant
E	Young's modulus
ν	Poisson's ratio
σ	Stress
α	Numerical constant for clamp configuration
β	Numerical constant for clamp configuration
Γ	Ratio of Planck's constant and Boltzmann constant
δ	Limit on temperature ratio
I	Moment of inertia
D	Diameter of CNTs
A	Area of cross section

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Chapter 1 Introduction

1.1 Quantum Computation and photons

During the last few decades, we have realized that by harnessing quantum mechanical effects, we can bring in extra power and functionality in the processing, encoding, and transferring of information [1]. There are several applications to it. Quantum key distribution offers secure communication [2]. Quantum Metrology gives us accurate measurements which could not be achieved without quantum mechanics [3]. Quantum lithography could make it possible to manufacture devices whose characteristic lengths are smaller than the wavelength of light [4]. The Quantum computer is the most anticipated and startling future quantum technology through which computation speed could be increased manifold [1, 5].

To develop quantum computation and other quantum mechanics applications, mastering manufacturing and designing nano-scale devices is essential. The fact that the role of quantum entanglement in quantum computing is not yet fully grasped does not make it any easier. As engineers, we must fabricate devices that can exploit these quantum mechanical effects to give us a deeper understanding of the fundamental working of nature.

Scalable physical qubits (two-state quantum systems) that can be used to implement logic gates and can be very well isolated from the environment are essential for realizing a quantum computer [6]. Including NMR, solid-state and super-conducting systems, cavity quantum electrodynamics, and many other implementations are the target of research [7]. Nevertheless, over the past decade, isolated particles of light, photons, have become the leading research approach in the industry. It is because they lack decoherence in the isolated photons, which adversely affects other systems, and because of their easy manipulation to realize logic gates.

1.2 Phononics

One of the crucial issues in the progress of the electronics industry is the heat removal through electronic circuits due to increasing levels of power density and speeds [8]. One of the significant issues in photonics and optoelectronics is self-heating [9].

Such facts instigated recent interest in the thermal properties of materials. The main heat carriers in a variety of electronic and material systems are acoustics phonons which are the fast-moving quanta of crystal lattice vibrations [10, 11, 12, 13, 14, 15, 16]. The phonon and thermal properties of nanostructures are substantially different from those of bulk crystals [17, 18, 19, 20, 21, 22]. Due to increased phonon-boundary scattering [17, 18], changes in the dispersion of phonon, and DOS [10-22], semiconductor thin films and nanowires do not conduct heat as well compared to bulk crystals.

Theoretical studies have suggested that phonon transport may show exotic behavior in two-dimensional and one-dimensional systems, leading to enormous values of intrinsic thermal conductivity [12, 13]. Such results have questioned the validity of Fourier's law in low-dimensional systems [23, 24] and have instigated further interest in acoustic phonon transportation in two-dimensional systems.

For the above-mentioned Phononics applications in quantum computation and phonon transportation, we require a single stream of antibunched particles. A phonon cannot enter a cavity when there is already a phonon present in that cavity. It is termed as Phonon blockade effect. By exploiting this effect, we might be able to produce antibunched phonon particles at feasible temperatures and length scale of the system.

1.3 Problem Statement

This work aims to achieve a higher limit on the temperature ratio (i.e., the ratio of surrounding temperature to characteristic system temperature due to vibrations) concerning the second-order correlation coefficient. The current limit, as proposed in [25], which corresponds with coupling two nano/micro mechanical resonators (i.e., two phonon modes), is very low. Using Carbon Nano Tubes as resonators, we can reach the Deci-Kelvin range, which is very low for commercial implementation and experimentation. This project aims to increase the temperature limit by coupling an array of nanomechanical resonators (CNTs) with each other and finding suitable system parameters for realizing the phonon blockade effect at around 1 Kelvin range.

1.4 Antibunching in mechanical resonators

With sufficiently high Kerr nonlinearity in an optical cavity, we can prevent more photons from entering the cavity once a photon is inside. It is called as phonon blockade effect. Light passing through the optical cavity shows strong antibunching, which is then used to develop a single stream of photons [26]. This same effect can be exhibited in phonons in a strong non-linear mechanical resonator [27, 28, 29, 30, 31, 32], but the intrinsic nonlinearity of mechanical resonators is not enough to observe it. To eliminate the problem of weak nonlinearity in optical cases, Liew and Savona found that photons exhibit strong antibunching in two coupled optical cavities with a weak optical nonlinearity.

In the paper by Shenggou Guan [25], they have discussed coupling a driven linear nano/micro mechanical resonator with one non-linear mechanical resonator to obtain strong antibunching. They found the optimal conditions at which antibunching is possible analytically and numerically by solving the Liouville-Von Neumann master equation. The analytical solution, including the temperature dependency that they have computed, is given below.

$$g^{(2)}(0) \simeq \frac{2\rho_{66}}{\rho_{44}^{2}}$$

$$\rho_{44} = \frac{F^{2}\,|\bar{\Delta}|^{2}}{|J^{2} - \bar{\Delta}^{2}|^{2}} + \eta_{th} \tag{1}$$

$$\rho_{66} = \frac{F^{4}|U\,(J^{2} + 2\bar{\Delta}^{2}) + 2\bar{\Delta}^{3}|^{2}}{2J^{8}\,|U + 2\bar{\Delta}|^{2}} + \frac{2F^{2}|\bar{\Delta}|^{2}}{J^{4}}\eta_{th} + \eta_{th}^{2}$$

In the above equation, $g^{(2)}(0)$ is the equal time second-order correlation coefficient, ρ is the density matrix, F is the force by which the resonator is driven, Δ is the detuning between the free vibration frequency of the oscillator and the driven frequency, U is the intrinsic Kerr nonlinearity of the non-linear oscillator, J is linear coupling strength, η_{th} is the average thermal phonon number of the mechanical resonators.

Their work forms a basis for studying antibunching in purely mechanical systems. They have studied the phonon antibunching effect in coupled mechanical resonators with weak intrinsic nonlinearity at zero and finite temperatures. In the weak forcing, analytical results were derived for the phonon blockade effect in a strong coupling regime.

Moreover, the effect of temperature on phonon antibunching was investigated based on the stationary Lindblad master equation. At zero temperatures, perfect phonon antibunching exists, but when thermal noise exists, we obtain a suppressed phonon antibunching effect. It is also observed that the temperature variations have no effect on the quantum correlation of the phonon mode when the surrounding temperature is present in the quantum regime. Also, the effect of forcing on phonon antibunching is discussed. A strong forcing is found to be more helpful in preserving the antibunching effect at finite temperatures.

In the research paper by Sai Subramaniam [33], we have seen that by using carbon-nanotubes as the mechanical resonators, we can achieve antibunching at near Kelvin temperatures, which is incredible as earlier works talk about antibunching at Milli Kelvin scale. Thus, exploration of such nano-systems is required to exploit their properties.

1.5 Carbon Nano Tubes

As explained by Iijima and Ichihashi [34, 35], carbon nanotubes are perfect systems for exploiting electromechanical effects. This is due to their small length, diameter, low mass, and structure with no defects [36]. Experimental reports on carbon nanotubes show auspicious high values for natural frequencies. This property allows them to be used in high-frequency nanomechanical devices [36, 37, 38]. For some of these devices, the frequency of oscillation is the main requirement [37].

According to [37], one of the effective methods to control the electronic charge and the spin quantum states is to couple an electromechanical resonator with carbon nanotubes. As they have stated, by doing so, we can separate coupled and electronically tunable mechanical resonators. The individual parameters, such as their frequencies, can be tuned by the bottom gates. Also, the conductance of the resonators can be modulated non-locally by phonon interactions between the two resonators [37]. Thus, such a strongly coupled carbon nanotube-based nanomechanical resonator array provides a platform for our study.

1.6 Non-local theory

The nanostructure's length scales are often sufficiently small; hence, for the applicability of classical continuum models, we need to consider the small length scales such as lattice spacing between individual atoms, grain size, etc.

Hence, Eringen’s non-local elasticity theory [39, 40] helps treat phenomena whose origins lie in regimes smaller than the classical continuum models.

This theory takes account of remote action forces between atoms. It causes the stresses to depend on the strains not only at an individual point under consideration but at all body points. In this theory, the internal size or scale could be represented in the constitutive equations simply as material parameters. Such non-local continuum mechanics has been widely accepted and has been applied to many problems, including wave propagation, dislocation, crack problems, etc.

For the non-local elasticity, a differential form exists for relating stress and strain values.

$$\boldsymbol{(1 - \xi^2 l^2 \nabla^2)\sigma_{kl}(\bar{x}) = \sigma^c_{kl}(\bar{x}) = C_{klmn}\varepsilon_{mn}(\bar{x})} \tag{2}$$

Where C_{ijkl} is the elastic modulus tensor of classical isotropic elasticity and ε_{ij} is the strain tensor, ∇^2 denotes the second order spatial gradient applied on the stress tensor σ_{kl} and $\xi = \frac{e_0 a}{L}$. Here $e_0 a$ is called the scale parameter. For our case, we will investigate the effect of this parameter on our environment temperature required for observing phonon antibunching.

Chapter 2 Quantum Analysis

2.1 System Modelling

Using Carbon Nanotubes as nanomechanical resonators, we may achieve antibunching at a near Kelvin temperature scale [33]. So, the system under consideration consists of an array of Carbon nanotubes arranged in a linear fashion equidistant from each other, as shown in Figure 2.1. The nanotubes are clamped on both sides with metal pads, and each of them is capacitively coupled with the gate having the voltage V_G.

The number of nanotubes, the material and geometric properties of the nanotubes, the separation and charges provided to the nanotubes, and the external actuation field becomes the tunable parameters of the system. We adjust the values of these parameters to achieve phonon blockade at a higher surrounding temperature.

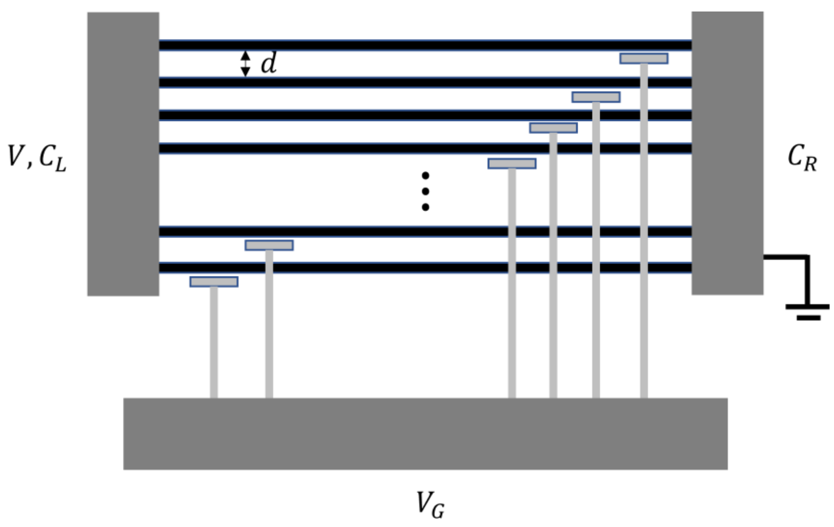


Figure 2.1 Schematic of our model

2.2 Hamiltonian for our proposed system

For the time being, all the resonators are considered non-linear, and all of them are harmonically driven by external forcing amplitude F with a frequency w_d. This type of Hamiltonian can be broken into four subparts, as shown in equation (3).

$$\boldsymbol{H = H_{free} + H_{nl} + H_{drive} + H_{co}} \tag{3}$$

In the above equation, H_{free} is the free Hamiltonian of all the resonators combined, H_{nl} is the Hamiltonian describing the K_{err} nonlinearity, U of all the resonators, and H_{co} Represents the coupled interaction between the resonators [41]. In our system, n number of mechanical nodes are being driven with respective frequencies and phase differences which is described by the term H_{co}. These terms are described in equation (4).

$$\begin{aligned} \boldsymbol{H_{free}} &= \boldsymbol{\omega_m \left(\sum_{i=1}^{n} b_i^\dagger b_i \right)} \\ \boldsymbol{H_{nl}} &= \boldsymbol{\sum_{i=1}^{n} U_i \left(b_i^\dagger + b_i \right)^4} \\ \boldsymbol{H_{drive}} &= \boldsymbol{\sum_{i=1}^{n} \Omega_i \left(b_i^\dagger e^{-\iota\omega_p t} + b_i e^{\iota\omega_p t} \right)} \\ \boldsymbol{H_{co}} &= \boldsymbol{\sum_{i=1}^{n} \sum_{j=1}^{n} J_{ij} \left(b_i^\dagger + b_i \right) \left(b_j^\dagger + b_j \right)} \end{aligned} \tag{4}$$

Where $b_i^\dagger$ and b_i are the creation and annihilation operators for the phonon mode of the i^{th} nanomechanical resonator with resonance frequency ω_m. U is the K_{err} nonlinearity for i^{th} resonator and J_{ij} is the coupling strength between i^{th} and j^{th} resonators. The term J_{ij} depends upon the potential energy (U_{ij}), masses (m_i, m_j) and resonant frequency (ω_m) by the following equation.

$$J_{ij} = \frac{1}{2}\frac{d^2U_{ij}^e}{dz^2}\sqrt{\frac{1}{m_1m_2\omega_m^2}} \tag{5}$$

Using rotating wave approximations, H_{nl} and H_{co} can be simplified into the following forms.

$$\begin{gathered} H_{nl} = \sum_{i=1}^{n} U_i\left(b_i^\dagger b_i^\dagger b_i b_i\right) \\ H_{co} = \sum_{i}^{n-1}\sum_{j=i+1}^{n} J_{ij}(b_i^\dagger b_j + b_j^\dagger b_i) \end{gathered} \tag{6}$$

For simplicity of equations, we consider that the resonators share the same frequency, Kerr nonlinearity, and the force amplitude. The net Hamiltonian for the system is as stated in the equation (7) where detuning $\Delta = (\omega_m - \omega_p)$.

$$\begin{gathered} H = \omega_m\left(\sum_{i=1}^{n} b_i^\dagger b_i\right) + U\left(\sum_{i=1}^{n} b_i^\dagger b_i^\dagger b_i b_i\right) + \mathrm{F}\sum_{i=1}^{n}\left(b_i^\dagger e^{-\iota\omega_p t} + b_i e^{\iota\omega_p t}\right) \\ + \sum_{i}^{n-1}\sum_{j=i+1}^{n} J_{ij}(b_i^\dagger b_j + b_j^\dagger b_i) \end{gathered} \tag{7}$$

2.3 Lindblad Master equation

The reduced density matrix of the system can be obtained using temperature dependent Lindblad master equation. The temperature dependence is important as the phonons are very sensitive to temperature variation.

$$\frac{\partial\hat{\rho}}{\partial t} = 0 = -i\left[\hat{H},\ \hat{\rho}\right] + \frac{\gamma}{2}\sum_{i=1}^{n}\{(\eta_{th}+1)D[b_i]\hat{\rho} + (\eta_{th}D\left[b_i^\dagger\right]\hat{\rho})\} \tag{8}$$

In the above equation, $\widehat{H}$ is the Hamiltonian operator of the system, γ is the decay rate, $D[b_i]\hat{\rho} = 2b_i\hat{\rho}b_i^\dagger - b_i^\dagger b_i\hat{\rho} - b_i b_i^\dagger\hat{\rho}$ is the Lindblad operator, and $\eta_{th} = \frac{1}{e^{\frac{T_0}{T}}-1}$ is the average phonon number.

In the above equation, $\hat{\rho}$ is the reduced density matrix and not the full density matrix. So, the time derivative would still depend upon the full density matrix as the above equation is not a closed-time evolution equation. Although we will be looking at only a steady state solution, so above equation is perfect for us. Therefore, the steady-state equation becomes as follows.

$$\boldsymbol{\mathfrak{L}\widehat{\rho} = 0} \tag{9}$$

Where $\boldsymbol{\mathfrak{L}}$ is the so-called Liouvillian super operator. We should note that the n is the number of resonance modes present in our system and we will have as many creation and annihilation operators.

The second-order correlation function for the first resonator can be described as follows.

$$\boldsymbol{g^{(2)}(0) = \frac{Tr\left(\widehat{b}^\dagger\widehat{b}^\dagger\widehat{b}\widehat{b}\widehat{\rho}_{ss}\right)}{Tr\left(\widehat{b}^\dagger\widehat{b}\widehat{\rho}_{ss}\right)^2}} \tag{10}$$

In the above equation $\hat{\rho}_{ss}$ is the steady-state density matrix and is obtained from solving the eigenvalue equation (9). Note that $\hat{\rho}_{ss}$ will be the eigen vector corresponding to zero eigenvalue.

2.4 Numerical Approach

For solving the steady state Lindblad master equation, we have followed the steps from [42] and [43] and modified the equations accordingly to accommodate multiple number of modes.

First of all, we shall vectorize the $\hat{\rho}$ matrix and rest of the terms in the equation (8).

$$\begin{aligned} \boldsymbol{vec}\left(\left[\widehat{\boldsymbol{H}}, \widehat{\boldsymbol{\rho}}\right]\right) &= \left(\boldsymbol{I} \otimes \boldsymbol{H} - \boldsymbol{H}^{\boldsymbol{T}} \otimes \boldsymbol{I}\right) * \boldsymbol{vec}(\widehat{\boldsymbol{\rho}}) \\ \boldsymbol{vec}(\boldsymbol{D}[\boldsymbol{b_i}]\widehat{\boldsymbol{\rho}}) &= \boldsymbol{vec}\left(\boldsymbol{2b_i}\widehat{\boldsymbol{\rho}}\boldsymbol{b_i^\dagger} - \boldsymbol{b_i^\dagger b_i}\widehat{\boldsymbol{\rho}} - \widehat{\boldsymbol{\rho}}\boldsymbol{b_i^\dagger b_i}\right) \\ &= \boldsymbol{2}\left(\boldsymbol{b_i^\dagger}\right)^{\boldsymbol{T}} \otimes \boldsymbol{b_i} - \boldsymbol{I} \otimes \left(\boldsymbol{b_i^\dagger b_i}\right) - \left(\boldsymbol{b_i^\dagger b_i}\right) \otimes \boldsymbol{I} \end{aligned} \tag{11}$$

The annihilation operator on truncated fock-state space for a single mode can be written as follows.

$$\left[\boldsymbol{b_{N_m}}\right]_{\boldsymbol{ij}} = \begin{cases} \sqrt{\boldsymbol{i}}\,, & \boldsymbol{j = i + 1} \\ \boldsymbol{0}, & \boldsymbol{otherwise} \end{cases} \tag{12}$$

We will use a postulate stated in [44] to separate the effects of different modes of the system. It states that the state-space of a composite system which includes n number of subsystems is the tensor or Kronecker product of state space of each component. Thus, for doing so, the n number of fock-states for the system can be written as follows.

$$\begin{gathered} \boldsymbol{b_1} = \boldsymbol{b_{N_m}} \otimes \boldsymbol{I_{N_m}} \otimes \boldsymbol{I_{N_m}} \otimes \ldots \otimes \boldsymbol{I_{N_m}} \\ \boldsymbol{b_2} = \boldsymbol{I_{N_m}} \otimes \boldsymbol{b_{N_m}} \otimes \boldsymbol{I_{N_m}} \otimes \ldots \otimes \boldsymbol{I_{N_m}} \\ \vdots \\ \boldsymbol{b_n} = \boldsymbol{I_{N_m}} \otimes \ldots \otimes \boldsymbol{I_{N_m}} \otimes \boldsymbol{I_{N_m}} \otimes \boldsymbol{b_{N_m}} \end{gathered} \tag{13}$$

Where, I_N is the $N \times N$ identity matrix.

We can obtain the Hamiltonian by doing simple matrix multiplications by substituting the above equation set back in equation (7). After that, we can substitute the obtained Hamiltonian and the fock state matrices in equation (11) to obtain Lindblad super-operator $\mathfrak{L}$. Finally, we reach to a linear set of equations (9), with an extra condition on the trace of the density matrix, which can be solved by setting one of the equations as $Tr(\rho) = 1$ in vector format. The resulting linear system of equations is inhomogeneous and can be solved by ad-hoc methods. In MATLAB, we can translate all the above steps into compact code.

The matrices that we have considered in the above steps are very sparse, for which we may use the sparse representation of the matrices present in MATLAB. For solving the set of linear equations, we have used the built-in Eigen solver, which provides us with the eigenvector corresponding to the eigen value whose absolute value is closest to zero.

The increase in the number of modes poses a computation problem as with each increase in the number of phonon modes, row, and column size of $\mathfrak{L}$ matrix each gets multiplied by a factor of N^{2n}. We have used three modes to obtain results and show that with an increase in the number of phonon modes, we may be able to increase the limit on temperature ratio for antibunching. Even for such a small number, this approach takes a few hours to carry out all the computation with 16 GB of RAM. An increase in efficiency and compact computation is highly recommended.

Chapter 3 Engineering Analysis

3.1 Natural Frequency for CNT

Carbon nanotubes may be modelled by non-local cylindrical shells. Hence, for the case of cylindrical shells, the non-local constitutive relations [10] in polar coordinate system (R, θ) are given by

$$\begin{aligned}
\boldsymbol{\sigma_{xx}} - (\boldsymbol{e_0 a})^2 \left[\frac{\partial^2 \boldsymbol{\sigma_{xx}}}{\partial \boldsymbol{x}^2} + \frac{\mathbf{1}}{\boldsymbol{R}^2}\frac{\partial^2 \boldsymbol{\sigma_{xx}}}{\partial \boldsymbol{\theta}^2}\right] &= \frac{\boldsymbol{E}}{\mathbf{1} - \boldsymbol{\nu}^2}(\boldsymbol{\varepsilon_{xx}} + \boldsymbol{\nu}\boldsymbol{\varepsilon_{\theta\theta}}) \\
\boldsymbol{\sigma_{\theta\theta}} - (\boldsymbol{e_0 a})^2 \left[\frac{\partial^2 \boldsymbol{\sigma_{\theta\theta}}}{\partial \boldsymbol{x}^2} + \frac{\mathbf{1}}{\boldsymbol{R}^2}\frac{\partial^2 \boldsymbol{\sigma_{\theta\theta}}}{\partial \boldsymbol{\theta}^2}\right] &= \frac{\boldsymbol{E}}{\mathbf{1} - \boldsymbol{\nu}^2}(\boldsymbol{\varepsilon_{\theta\theta}} + \boldsymbol{\nu}\boldsymbol{\varepsilon_{xx}}) \qquad (14) \\
\boldsymbol{\tau_{x\theta}} - (\boldsymbol{e_0 a})^2 \left[\frac{\partial^2 \boldsymbol{\tau_{x\theta}}}{\partial \boldsymbol{x}^2} + \frac{\mathbf{1}}{\boldsymbol{R}^2}\frac{\partial^2 \boldsymbol{\tau_{x\theta}}}{\partial \boldsymbol{\theta}^2}\right] &= \frac{\boldsymbol{E}}{\mathbf{2}(\mathbf{1} + \boldsymbol{\nu})}\boldsymbol{\gamma_{x\theta}}
\end{aligned}$$

For the present analysis the CNT is modelled as a Euler-Bernoulli beam. By solving the above constitutive relations, for a doubly clamped carbon nanorod, the natural frequency for the first mode can be written as follows:

$$\begin{gathered}
f = \frac{1}{2\pi} \times 22.373 \times \sqrt{\frac{1}{\lambda}} \times \sqrt{\frac{EI}{\rho A L^4}} \\
\lambda = 1 + 4\left(\frac{e_0 a}{L}\right)^2 \pi^2
\end{gathered} \qquad (15)$$

We may introduce several non-dimensional parameters to convert the above equation in a multivariable optimization problem.

$$K_1 = \frac{D}{L}, K_2 = \frac{e_0 a}{L}, K_3 = \frac{I}{D^4}, K_4 = \frac{A}{D^2} \qquad (16)$$

Where, D is the diameter of the carbon nanorod and the critical dimension for our system.

Also, for carbon nanorods, we may assume some standard values for the material properties as mentioned in [36, 45, 46, 47]. For our analysis, we have taken

$$E = 1\,TPa;\ \rho = 1400\,kg.m^{-3} \tag{17}$$

Substituting in the above equation we get,

$$f.D = \alpha \times \frac{K_1^2}{\sqrt{1+\beta.K_2^2}} \times \sqrt{\frac{K_3}{K_4}}$$
$$\alpha = 9.517 \times 10^4;\ \beta = 39.478 \tag{18}$$

3.2 Temperature-frequency relation

From the quantum analysis, we establish the value of the temperature ratio, $\frac{T}{T_0}$ (let's call it δ), for which we may observe phonon antibunching. Since T_0 is a characteristic temperature of the system, it's expression can be found in terms of the properties of the system [25].

$$T_0 = \frac{hf}{K_B} \tag{19}$$

This allows us to relate the system temperature with the system characteristic frequency. Larger the natural frequency of the resonator, larger is the system temperature that it can work at.

Thus, substituting back in the equation for T, we get,

$$T = 3.4815 \times 10^{-10} \times \delta \times f \tag{20}$$

Where, δ is the value of limit on temperature ratio.

After substituting the equation (20) in the equation (18), we get

$$\frac{T}{T_0}.D = \alpha \times \Gamma \times \delta \times \frac{K_1^2}{\sqrt{1+\beta.K_2^2}} \times \sqrt{\frac{K_3}{K_4}} \tag{21}$$
$$\Gamma = 3.4815 \times 10^{-10} \text{ second}^{-1}$$

The above equation is multi variable optimization problem, which we can use to understand the effects of different sizes and configurations of carbon nanorods. Note that in the above equation the values of α, and β changes with the different configurations of the resonators, i.e. clamped-clamped, simply supported, or cantilever. For all these the values of α and β are as follows:

Configuration	$\boldsymbol{\alpha}$	$\boldsymbol{\beta}$
Simply Supported	4.198×10^4	9.869
Doubly clamped	9.517×10^4	39.478
Cantilever	1.4954×10^4	2.467

Table 3.1 Variation of α and β with change in clamping configuration of resonators

Considering the single walled CNT as hollow cylindrical beam,

$$K_3 = \frac{I}{D^4} = \frac{\pi}{4D^4}\left(\frac{D}{2}+\frac{t}{2}\right)^4 - \frac{\pi}{4D^4}\left(\frac{D}{2}-\frac{t}{2}\right)^4 = \frac{\pi}{8}\left\{\left(\frac{t}{D}\right)+\left(\frac{t}{D}\right)^3\right\};$$
$$K_4 = \frac{A}{D^2} = \frac{\pi t}{D}$$
$$Let, K_5 = \sqrt{\frac{K_3}{K_4}} = \frac{1}{2}\sqrt{1+\left(\frac{t}{D}\right)^2} \tag{22}$$
$$Hence, T = \frac{\alpha \times \Gamma \times \delta}{D} \times \frac{K_1^2 K_5}{\sqrt{1+\beta.K_2^2}}$$

Chapter 4 Results & Discussion

4.1 Simulation Results

The local minima that we will get for some system hyperparameters values will depend on the system's configuration. Our system consists of an array of nanomechanical resonators, and we may apply some force to a set of these resonators and choose to have a set of them as non-linear. Based on this approach, we have chosen three different configurations for performing quantum simulations to find the best minima. Also, we will look at results pertaining to an array of three resonators.

For validation of results, we are comparing the analytical and numerical solution for two coupled resonators, one of which is linear and the other non-linear.

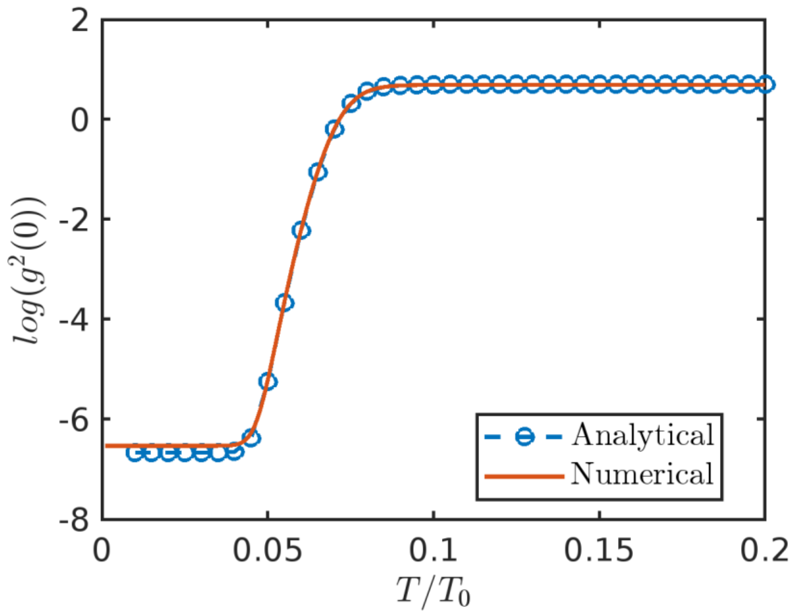


Figure 4.1 Comparison of analytical and numerical solution for n=2

In Figure 4.1, the results have been obtained by following the analytical solution mentioned in the literature review and the numerical approach mentioned in the previous section. For obtaining above results $F = 1\gamma, J = 10\gamma, U = 10^{-3}\gamma, \Delta = 0.29\gamma, and\ \gamma = 1.$

4.1.1 Configuration 1

All the resonators are non-linear and have equal forcing. The quantum simulation results are illustrated in Figure 5.1.

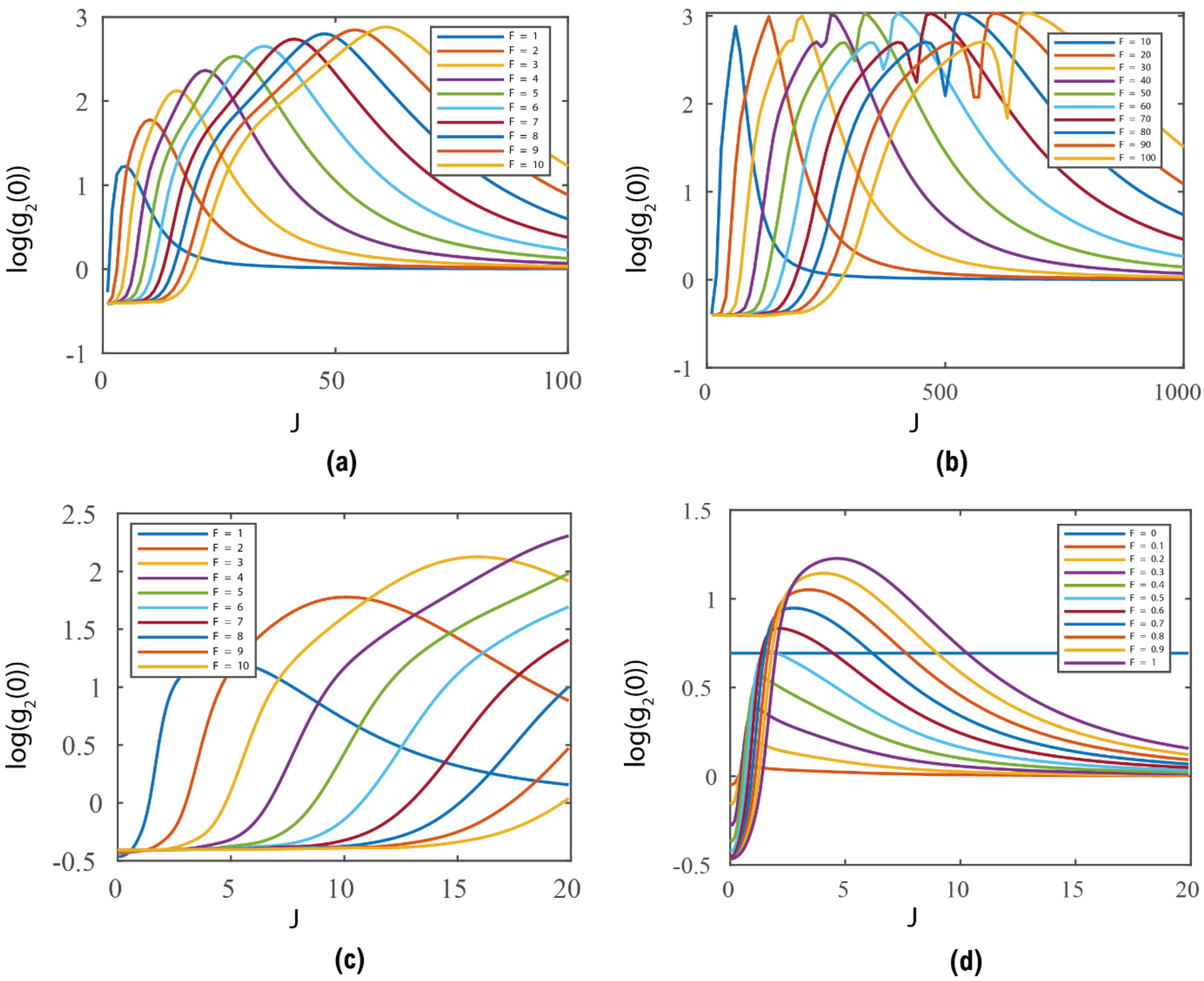


Figure 4.2(a) log(g2(0)) values for moderate forcing and moderate coupling regime. (b) log(g2(0)) values for strong forcing and strong coupling regime. (c) log(g2(0)) values for moderate forcing and weak coupling regime. (d) log(g2(0)) values for weak forcing and weak coupling regime.

The above sensitivity test is done at a detuning = 0.29 and U = 1E-3. We get two minima according to the above graphs. One in the weak coupling and weak forcing regime (Figure 5.1) and one in the strong coupling and strong forcing regime (Figure 5.1(b)). Both the minima seem unsuitable as the $g_2(0)$ values are very high in the strong coupling region, and the coupling is nearly zero in the weak coupling region.

4.1.2 Configuration 2

One of the resonators is linear and has forcing and the rest are non-linear with no forcing. The quantum simulation results are illustrated in Figure 5.2.

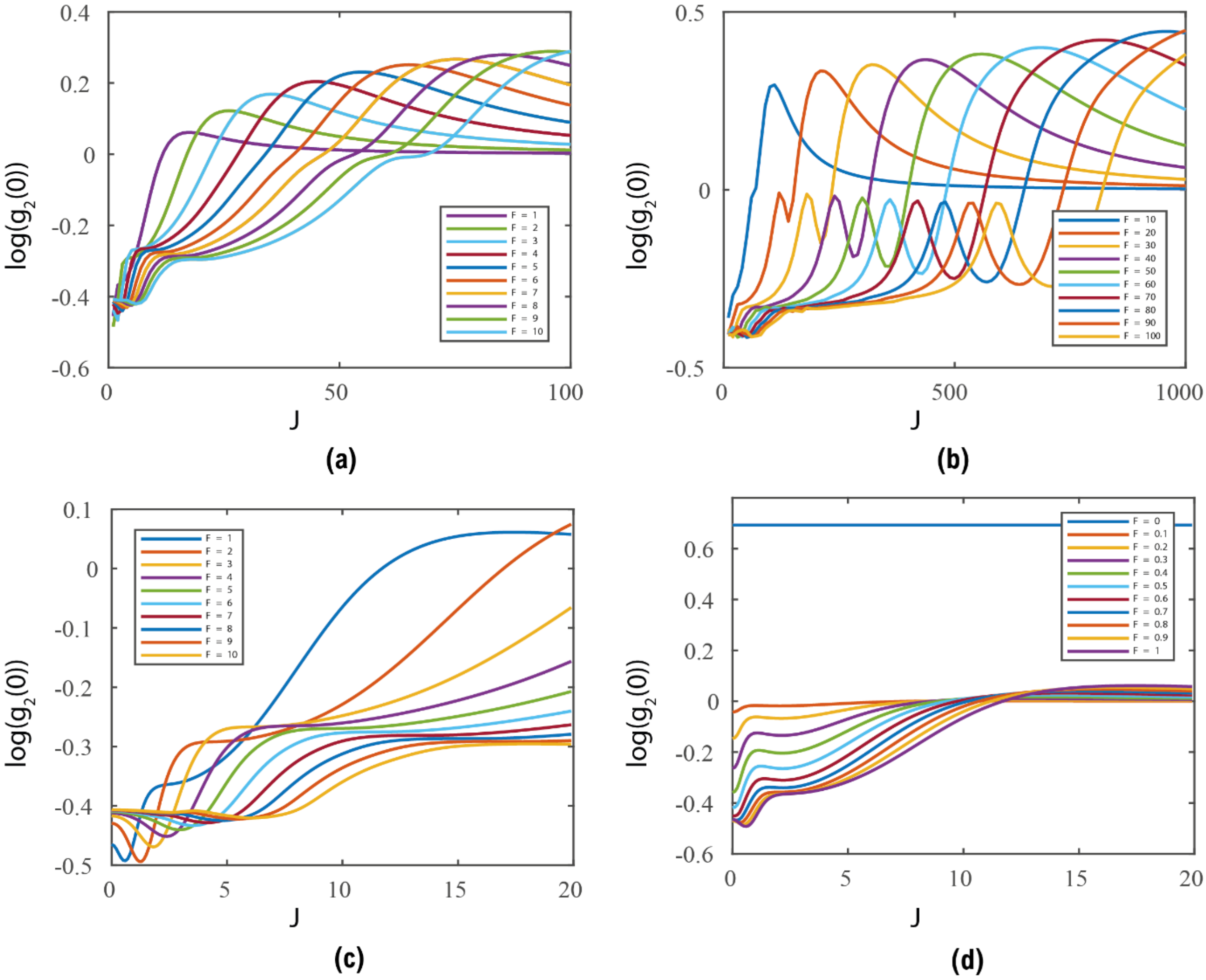


Figure 4.3 (a) log(g2(0)) values for moderate forcing and moderate coupling regime. (b) log(g2(0)) values for strong forcing and strong coupling regime. (c) log(g2(0)) values for moderate forcing and weak coupling regime. (d) log(g2(0)) values for weak forcing and weak coupling regime.

The above sensitivity test is done at a detuning = 0.29 and U = 1E-3. We get two minima according to the above graphs. One in the weak coupling regime and at $F = 2$ (Figure 4.2(c)) and one in the strong coupling and strong forcing regime (Figure 4.1(b)). We would get moderate antibunching for both the cases, so for both cases we develop $\log\,(g_2(0))$ vs temperature ratio T/T_0 curves to look get the maximum value of the temperature ratio for which antibunching may be possible.

4.1.3 Configuration 3

Two of the resonators are linear and have forcing and the remaining is non-linear with no forcing. The quantum simulation results are illustrated in Figure 4.4.

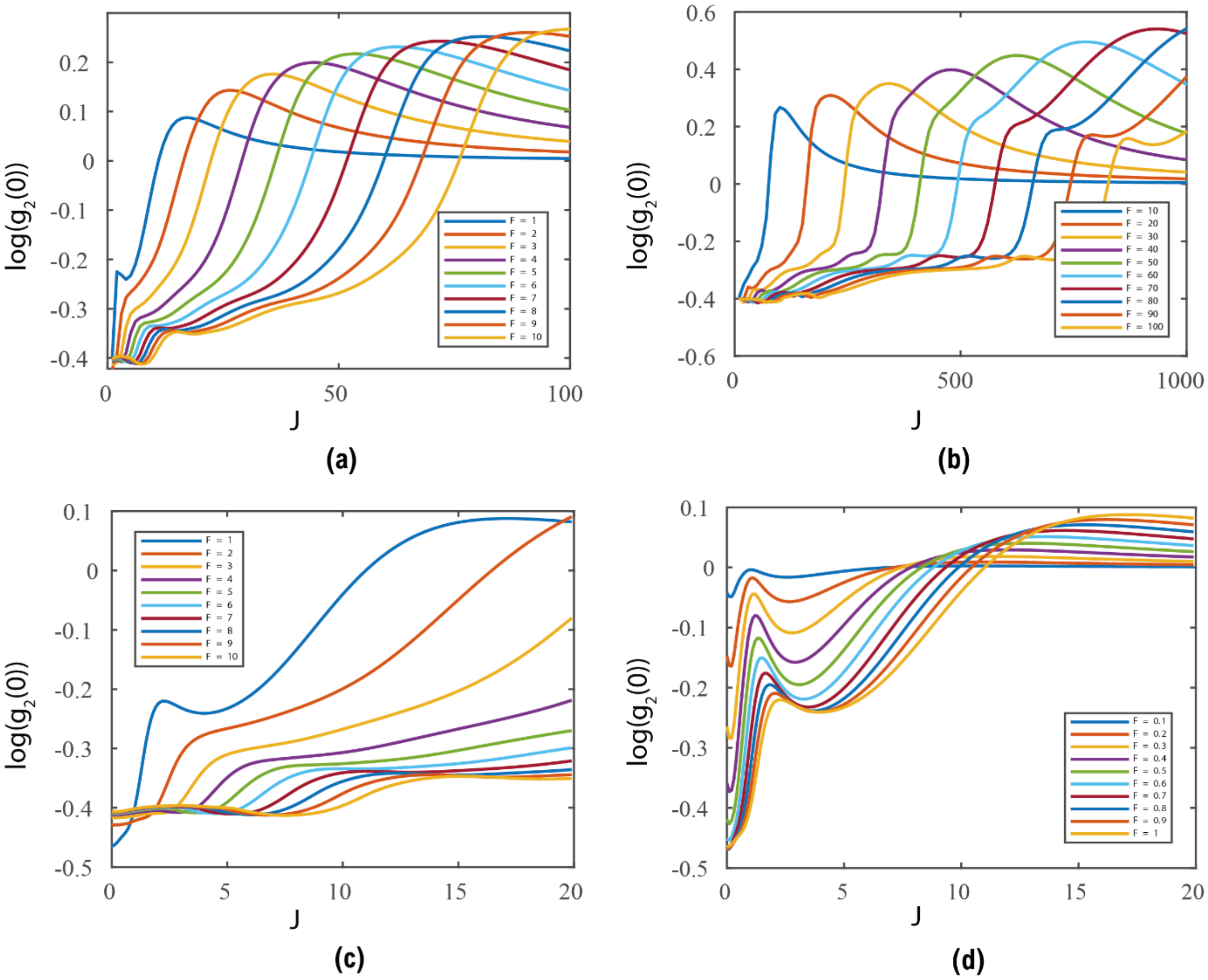


Figure 4.4 (a) log(g2(0)) values for moderate forcing and moderate coupling regime. (b) log(g2(0)) values for strong forcing and strong coupling regime. (c) log(g2(0)) values for moderate forcing and weak coupling regime. (d) log(g2(0)) values for weak forcing and weak coupling regime.

We get three minima according to the above graphs. One at $J = 100$ and $F = 100$ (Figure 4.3(b)), one in the extremely strong coupling and strong forcing regime (Figure 4.3(b)), and one at nearly zero coupling and weak forcing regime (Figure 4.3(d). The minima in extremely strong coupling regime and nearly zero coupling regime are not useful as stated in configuration 1. We may explore the minima near $J = 100$ and $F = 100$.

4.2 $g^{(2)}(0)$ vs Temp. ratio curves

4.2.1 For configuration 2 at F = 1.5 and J = 1

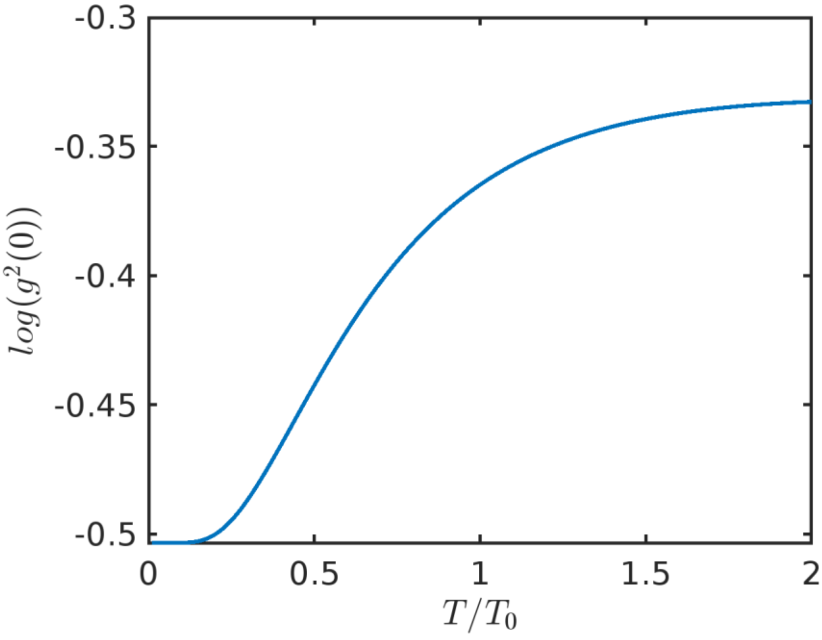


Figure 4.5 Plot between $\log(g_2(0))$ *vs* T/T_0

The minimum value stays for $\frac{T}{T_0} = 0.15$ according to the above plot.

4.2.2 For configuration 2 at F = 100 and strong coupling regime

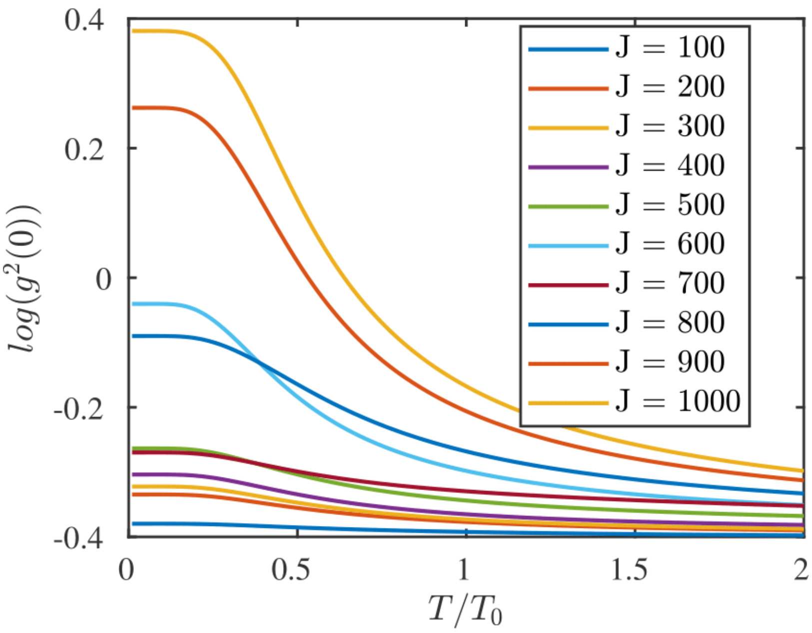
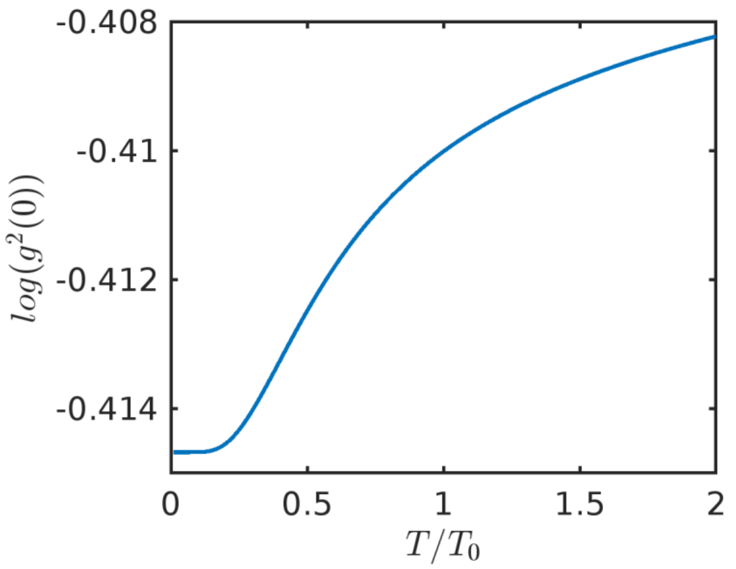


Figure 4.6 Plot between $\log(g_2(0))$ *vs* T/T_0

Similar curve characteristics are also observed when we have only two resonators at very high coupling regime. At moderate coupling we do not observe the transition of $g^{(2)}(0)$ values with change in Temperature ratios.

4.2.3 For configuration 3 at F = 90 and J = 64



The minimum value stays for $\frac{T}{T_0} = 0.13$ according to the above plot.

With two resonators, at J = 20 and F = 1, the maximum T/T_0 value for which we can observe antibunching was found to be **0.04**. Moreover, with increasing one resonator in series, we get the ratio for minima to be **0.15**. Following this approach, the surrounding temperature can be increased by around **five** times.

Now, since we have looked at the temperature ratio, we can use it to calculate the actual surrounding temperatures at which we get the minima in $g^{(2)}(0)$ values by substituting back in the equation (22).

4.3 Temperature-Length curve

For our analysis, we will consider two cases of diameters of SWCNTs (Single-Walled Carbon Nanotubes), $D = 0.678\ nm$ ((5,5) SWCNT), and $D = 1.356\ nm$ ((10,10) SWCNT). The nanotube length will vary from 5 nm to 50 nm, and the non-local parameter, e will be kept at 10 [46]. Based on the literature review [36, 45, 46, 47], it is assumed for the single element representing $C - C$ bond: the bond length $a = 0.142\ nm$, the stiffness $E = 1000\ GPa$, the density $\rho = 1400\ kg/m^3$ and the bond thickness $t = 0.34\ nm$.

For the clamped-clamped case, we get the following results.

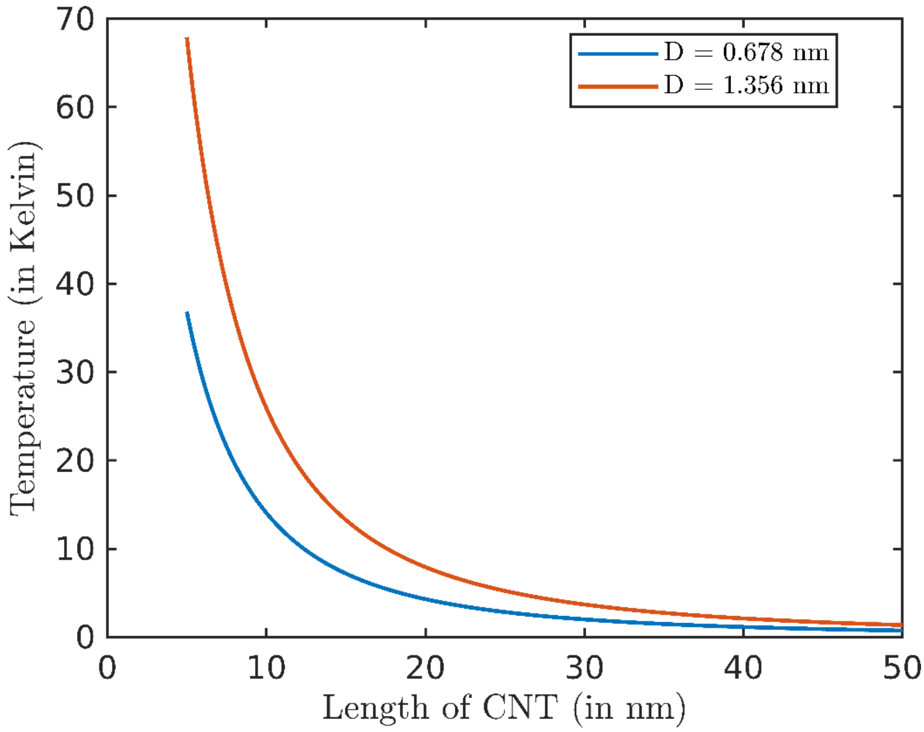


Figure 4.7 Variation of Surrounding temperature with the length of carbon nano tube.

From Figure 4.7 we can observe that we would achieve higher temperatures with a greater diameter. In India, the reasonable length of nanotube that can be manufactured is 50 nm, so the temperature values at which phonon blockade might be possible turn out to be around 1.3 Kelvin for three coupled resonators. This is quite an increase from the temperatures stated in [33], and [43].

The temperature values (in Kelvin) for the above case and other cases have been tabulated below.

	Clamped-Clamped		Simply Supported		Cantilever	
Length of nanotube (in nm) →	**25**	**50**	**25**	**50**	**25**	**50**
$\boldsymbol{D = 0.678\,nm}$	2.0073	0.7422	0.9137	0.3312	0.3282	0.1183
$\boldsymbol{D = 1.356\,nm}$	3.6998	1.3680	1.6841	0.6105	0.6048	0.2181

Table 4.1 System Temperature values for different diameters and lengths of CNTs

4.4 Conclusion

The results indicate that for the specified configurations, we observe minima for $U = 1 \times 10^{-3}\gamma$, $\Delta = 0.29\gamma$, for which $g^{(2)}(0)$ is close to zero. For these minima or any other minima that might exist in the region for which we may get strong antibunching, we have discovered the new temperature ratio (system temperature divided by intrinsic temperature) as 0.15 for three coupled nanoresonators. For two resonators, this limit was 0.04. Through this work, we have shown that coupling multiple nanomechanical resonators can increase the conventional limit on Temperature and push the temperatures to 1.3 Kelvin at 50nm length scales.

Chapter 5 References

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